

Relevance of PNPLA3, TM6SF2, HSD17B13, and GCKR Variants to MASLD Severity in an Egyptian Population.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly known as non-alcoholic fatty liver disease (NAFLD), is a frequent clinical condition globally. Single nucleotide polymorphisms (SNPs) associated with NAFLD have been proposed in the literature and based on bioinformatic screening. The association between NAFLD and genetic variants in Egyptians is still unclear. Hence, we sought to investigate the association of some genetic variants with NAFLD in Egyptians. Egyptians have been categorized into either the MASLD group (n = 205) or the healthy control group (n = 187). The severity of hepatic steatosis and liver fibrosis was assessed by a Fibroscan device. TaqMan-based genotyping assays were employed to explore the association of selected SNPs with MASLD. PNPLA3 rs738409 C>G variant is associated with the presence of MASLD with liver fibrosis, the severity of both hepatic steatosis and liver fibrosis, increased systolic and diastolic blood pressure and increased alanine aminotransferase (all $p < 0.05$), while the TM6SF2 rs58542926 C>T, HSD17B13 rs9992651 G>A, and GCKR rs1260326 T>C variants were not (all $p > 0.05$). The TM6SF2 rs58542926 T allele is associated with increased fasting blood glucose and a decreased waist circumference. The GCKR rs1260326 C allele is associated with decreased aspartate transaminase and diastolic blood pressure (all $p < 0.05$). Only after adjusting for the risk factors (age, sex, BMI, WC, HDL, TG, diabetes mellitus, and hypertension) F2 liver fibrosis score is negatively correlated with the HSD17B13 rs9992651 GA genotype. This study offers evidence for the association of the PNPLA3 rs738409 C>G variant with MASLD among Egyptians and for the association of the PNPLA3 rs738409 G allele, the TM6SF2 rs58542926 T allele, and the GCKR rs1260326 C allele with some parameters of cardiometabolic criteria.

Natural history and progression of metabolic dysfunction-associated steatotic liver disease.

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The natural history of metabolic dysfunction-associated steatotic liver disease (MASLD), previously referred to as non-alcoholic fatty liver disease (NAFLD), is complex and long. A minority of patients develop inflammation and risk progressive fibrosis that can result in cirrhosis. Progression to cirrhosis occurs in 3-5% of patients and often takes more than 20 years. This narrative review presents an update on the natural history of MASLD, discussing studies and risk estimates for progression to severe outcomes, such as decompensated cirrhosis or hepatocellular carcinoma. We highlight the dynamic progression of liver damage, how to identify patients whose disease progresses over time, and how risk factors might be mitigated to reduce the risk for disease progression.

The double roles of T cell-mediated immune response in the progression of MASLD.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly known as non-alcoholic fatty liver disease (NAFLD), has become a major cause of chronic liver disease and a significant risk factor for hepatocellular carcinoma, which poses a huge burden on global public health and economy. MASLD

includes steatotic liver disease, steatohepatitis, and cirrhosis, and the latter two cause great harm to human health and life, even complicated with liver cancer. Immunologic mechanism plays a major role in promoting its development into hepatitis and cirrhosis. Now more and more evidences show that T cells play an important role in the progression of MASLD. In this review, we discuss the double roles of T cells in MASLD from the perspective of T cell response pathways, as well as new evidences regarding the possible application of immunomodulatory therapy in MASH.

Non-invasive testing in metabolic dysfunction-associated steatotic liver disease.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), previously referred to as non-alcoholic fatty liver disease (NAFLD), is a leading cause of chronic liver disease, affecting up to 30% of the global population. MASLD is strongly associated with metabolic risk factors such as obesity and type 2 diabetes, and can progress to advanced stages including cirrhosis and hepatocellular carcinoma. Early diagnosis and accurate staging of fibrosis are critical in managing the disease and preventing complications. While liver biopsy has long been considered the gold standard for assessing fibrosis, it is invasive and carries associated risks. In response, non-invasive tests (NITs) have emerged as essential alternatives for the diagnosis and monitoring of MASLD. Key methods include blood-based biomarkers such as the Fibrosis-4 (FIB-4) score, NAFLD Fibrosis Score (NFS), and Enhanced Liver Fibrosis (ELF) test, as well as imaging modalities like vibration-controlled transient elastography (VCTE) and magnetic resonance elastography (MRE). These tests provide safer, more accessible methods for identifying liver fibrosis and guiding clinical management. They are integral in assessing disease severity, guiding treatment decisions, and monitoring disease progression, particularly in light of emerging therapies. NITs have become increasingly recommended by clinical guidelines as they reduce the need for invasive procedures like liver biopsy, improving patient care and outcomes. In conclusion, non-invasive testing plays a crucial role in the effective management of MASLD, offering reliable alternatives for diagnosis and monitoring while minimizing risks associated with traditional invasive methods.

Protocol for a Longitudinal Cohort Study to Understand Characteristics and Risk Factors Underlying Vibration-Controlled Transient Elastography-Diagnosed Metabolic Dysfunction-Associated Fatty Liver Disease Children.

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Metabolic-associated steatotic liver disease (MASLD) is a novel term proposed in 2023 to replace non-alcoholic fatty liver disease (NAFLD) with the aim of better reflecting its pathogenesis and clinical manifestations. Vibration-controlled transient elastography (VCTE) is an evidence-based, non-invasive imaging device used to evaluate liver fat deposition and fibrosis. It can effectively detect liver fat infiltration greater than 5%, which is much higher than the previous ultrasound detection rate (it is difficult to detect liver fat deposition below 30%). Nevertheless, the prevalence and characteristics of MASLD children diagnosed based on these updated criteria are currently not well established. Currently, a prospective multi-center population-based cohort study is being conducted in Wuxi, China, spanning from 2023 to 2035, involving 5600 children from four primary schools. Throughout the study's baseline and follow-up periods, yearly physical examinations, laboratory tests, VCTE assessments, and

bioelectrical impedance analysis are being conducted to measure MASLD-related biomarkers. Additionally, a questionnaire is being administered to inquire about dietary habits. MASLD is being diagnosed based on clinical and laboratory criteria, and the corresponding prevalence is being assessed. Recruitment began in March 13, 2023. To date, 1475 participants have completed the physical examination and questionnaire survey. Our study investigated the prevalence of MASLD and its influencing factors in Chinese school-age children and adolescents. By collecting and analyzing data from physical examinations and survey questionnaires, it may propose new avenues for guiding the treatment and early-stage prevention of MASLD in children. Chinese Clinical Trials Registry (NO. ChiCTR2400080508).

The Link Between Metabolic Dysfunction-Associated Steatotic Liver Disease and Gastroesophageal Reflux Disease.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly nonalcoholic fatty liver disease (NAFLD), and gastroesophageal reflux disease (GERD) are prevalent chronic conditions with escalating global incidence. This study delves into the intricate interplay between MASLD and GERD. The primary objective is to comprehensively explore the association between MASLD and GERD, investigating how various factors contribute to the coexistence and potential exacerbation of these conditions. We conducted a literature search in PubMed and Google Scholar of only human studies over the past 10 years. The search included systematic review, meta-analysis, editorial, and cross-sectional studies of patients with MASLD and GERD. The prevalence of GERD in patients with MASLD was higher, with various risk factors coming into play. Obesity was identified as an independent risk factor for both GERD and MASLD. However, obese patients predominantly had higher disease progression. Lifestyle factors like physical activity and dietary modifications emerge as promising strategies to mitigate risk.

Natural history of metabolic dysfunction-associated steatotic liver disease.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), which has been the term for non-alcoholic fatty liver disease (NAFLD) since June 2023, represents the most common liver disease worldwide and is a leading cause of liver-related morbidity and mortality. A thorough knowledge of the disease's natural history is required to promptly stratify patients' risks, since MASLD is a multifaceted disorder with a broad range of clinical phenotypes. The histological disease spectrum ranges from isolated hepatic steatosis, currently named as metabolic dysfunction-associated steatotic liver (MASL), to metabolic dysfunction-associated steatohepatitis (MASH) and eventually may accumulate hepatic fibrosis and develop cirrhosis and/or hepatocellular carcinoma (HCC). Several risk factors for fibrosis progression have been identified, while the disease's progression displays notable dynamism and bidirectionality. When compared to the general population, all MASLD histological stages are substantially related with greater overall mortality, and this association exhibits a disease severity-dependent pattern. Interestingly, the fibrosis stage is the most accurate predictor of mortality among MASLD patients. The mortality attributed to MASLD predominantly stems from issues linked with the liver and cardiovascular system, as well as HCC and extrahepatic cancers. In light of the disease natural course, it is crucial to

prioritize the identification of at-risk patients for disease progression in order to effectively address and change modifiable risk factors, hence mitigating disease complications. Further investigation is required to define the phenotype of rapid progressors more precisely as well as to improve risk stratification for HCC in non-cirrhotic individuals.

From NAFLD to MASLD: Meta-analysis and systematic review of NAFLD patients in Türkiye in terms of metabolic profile and MASLD potential.

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Non-alcoholic Fatty Liver Disease (NAFLD) is both a cause and a consequence of metabolic disturbances. Consequently, the disease term has recently changed to Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD). Türkiye is one of the leading countries with high incidences of diseases such as diabetes, obesity, metabolic syndrome, and fatty liver. This study aims to identify the metabolic parameters and MASLD potential of NAFLD in Türkiye. All NAFLD studies conducted in Türkiye were systematically searched using the keywords "fatty liver disease" AND "Türkiye" on PubMed, Scopus, and Web of Science databases. A total of 2653 articles were scanned, and 120 studies were eligible for meta-analysis. The metabolic parameters were meta-analyzed from a broad perspective. According to the meta-analysis results, there were significant increases in waist circumferences (mean difference: 10.90, $p < 0.00001$), HOMA-IR (mean difference: 2.13, $p < 0.00001$), aspartate aminotransferase (AST) (mean difference: 17.82, $p < 0.00001$), systolic blood pressure (SBP) (mean difference: 5.86, $p < 0.00001$), and C-reactive protein (CRP) levels (mean difference: 0.95, $p < 0.00001$). These parameters are representative biochemical findings of disturbed glucose metabolism, lipid profile, blood pressure, and acute phase response mechanisms. Furthermore, the analysis of all related parameters commonly found among the articles confirmed these metabolic dysfunctions. NAFLD is a metabolic disease that encompasses multiple pathways related to glucose and lipid metabolism, vascular function, inflammation, and acute phase responses. Additionally, our results suggest that Turkish NAFLD patients identified in previous studies mostly have MASLD. This is the first meta-analysis study indicating changes in metabolism-related parameters with a cumulative meta-analysis of all Turkish NAFLD studies.

Thyroid Hormone and Mitochondrial Dysfunction: Therapeutic Implications for Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD).

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Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly termed nonalcoholic fatty liver disease (NAFLD), is a widespread global health concern that affects around 25% of the global population. Its influence is expanding, and it is anticipated to overtake alcohol as the leading cause of liver failure and liver-related death worldwide. Unfortunately, there are no approved therapies for MASLD; as such, national and international regulatory health agencies undertook strategies and action plans designed to expedite the development of drugs for treatment of MASLD. A sedentary lifestyle and an unhealthy diet intake are important risk factors. Western countries have a greater estimated prevalence of MASLD partly due to lifestyle habits. Mitochondrial dysfunction is strongly linked to the development of MASLD. Further, it has been speculated that mitophagy, a type of mitochondrial quality control, may be impaired in MASLD. Thyroid hormone (TH) coordinates signals from the nuclear and mitochondrial genomes to

control mitochondrial biogenesis and function in hepatocytes. Mitochondria are known TH targets, and preclinical and clinical studies suggest that TH, thyroid receptor β (TR- β) analogs, and synthetic analogs specific to the liver could be of therapeutic benefit in treating MASLD. In this review, we highlight how mitochondrial dysfunction contributes to development of MASLD, and how understanding the role of TH in improving mitochondrial function paved the way for innovative drug development programs of TH-based therapies targeting MASLD.

New Therapeutic Approaches for the Treatment of Patients with Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD) and Increased Cardiovascular Risk.

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Metabolic dysfunction-associated steatotic liver disease (MASLD) was previously known as nonalcoholic fatty liver disease (NAFLD). The main characteristic of the disease is the process of long-term liver inflammation, which leads to hepatocyte damage followed by liver fibrosis and eventually cirrhosis. Additionally, these patients are at a greater risk for developing cardiovascular diseases (CVD). They have several pathophysiological mechanisms in common, primarily lipid metabolism disorders and lipotoxicity. Lipotoxicity is a factor that leads to the occurrence of heart disease and the occurrence and progression of atherosclerosis. Atherosclerosis, as a multifactorial disease, is one of the predominant risk factors for the development of ischemic heart disease. Therefore, CVD are one of the most significant carriers of mortality in patients with metabolic syndrome. So far, no pharmacotherapy has been established for the treatment of MASLD, but patients are advised to reduce their body weight and change their lifestyle. In recent years, several trials of different drugs, whose basic therapeutic indications include other diseases, have been conducted. Because it has been concluded that they can have beneficial effects in the treatment of these conditions as well, in this paper, the most significant results of these studies will be presented.
